Message Intelligence System — Theory Concepts

1. Conditional Probability

Probability of event A given that event B has already occurred: **P(A|B) = P(A ∩ B) / P(B)**, P(B) > 0.

> Example: P(Spam | message contains URL) = (spam messages with URL) / (all messages with URL).

2. Bayes' Theorem

P(C|X) = P(X|C) · P(C) / P(X)

Term	Name	Meaning in spam detection
P(C)	Prior	How common spam is overall (≈18.7%)
P(X|C)	Likelihood	How typical these features are for class C
P(X)	Evidence	Total probability of seeing these features (normaliser)
P(C|X)	Posterior	Probability the message is spam/legit given its features

Importance: it converts "how likely are these features if the message is spam" (easy to learn) into "how likely is spam given the features" (what we need). The predicted class is the one with the highest posterior.

3. Naive Bayes assumptions

- **Conditional independence** of features given the class: $P(x_1 \ldots x_n|C) = \prod P(x_i|C)$
- All features contribute equally
- Gaussian NB: each numeric feature is normally distributed within each class
- Training data represents real data

4. K-Nearest Neighbors & Support Vector Machine

KNN — lazy learner: stores the data; for a new message finds the K closest training points (Euclidean / Manhattan / cosine distance) and takes a majority vote. Needs feature scaling. Small K → overfits, large K → underfits.

SVM — finds the hyperplane with the **maximum margin** between classes; only **support vectors** define it. Margin width = 2/‖w‖. **C** controls the penalty for margin violations; **kernels** (linear, RBF, polynomial) allow non-linear boundaries; **gamma** controls the reach of a single point in RBF.

5. Distance-based vs Probabilistic vs Margin-based

	KNN (distance)	Naive Bayes (probabilistic)	SVM (margin)
Idea	Similar points share labels	Bayes' theorem + class probabilities	Maximum-margin boundary
Training	None (lazy)	Very fast	Slower (optimisation)
Prediction	Slow	Very fast	Fast
Scaling needed	Yes	No (helps)	Yes
Output	Vote ratio	True probability	Distance from boundary
Weakness	Noise, many features	Independence assumption	Tuning C/gamma

6. Evaluation metrics

Metric	Formula	Meaning
Accuracy	(TP+TN)/all	Overall correct (misleading on imbalanced data)
Precision	TP/(TP+FP)	Of messages flagged spam, how many really are spam
Recall	TP/(TP+FN)	Of all real spam, how much we caught
F1	2PR/(P+R)	Balance of precision and recall